Milestone 3

Remari Hudson, Ryan Schultz, Mariah Weldon, Lance West, Malia Mitchell

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The user interface to be created for the Automated Recall Management System (ARMS) will focus more on simplicity, clarity, and basic functionality to ensure effective tracking and communication during the recall process. A header section will obviously contain the company logo and navigation menu while a main dashboard will show total notifications, number of responses received, number of responses pending action(s), and the number of pumps replaced. Notification progress will show a visual representation of notifications sent via email or text and a real-time update of notifications sent per minute.

The next major section being notifications will display notification logs which will include the customer's name, contact information, date and time of notification sent and its delivery method. This section will also contain search and filter bars to further simplify the user process of finding specific cases based on certain parameters. The last part of this section will include options to resend notifications to customers who have yet to respond to the initial attempt at contacting.

The next major section implemented into the UI will be in the responses section. This will first include the response logs which is a table displaying the customer's name, response status (acknowledged, pending), date and time of response recorded, and the form completion status. The only other major section here will be the response tracking which will have color coding as a visual representation of response status. Along with that will be the ability to drill down into individual responses for certain details.

Obviously, our interface will next have to include a section dedicated to settings. This will first include user management settings. For example, user management will allow users to add/remove other users based on different access levels (admin, manager, operator). There will be a notification settings section dedicated to configuring notification templates for email and text. This could also be used to set up automated reminders for non-responders. Finally, this section will include system alerts which will receive alerts for system errors or delays in notification delivery.

The next two sections will be for prototype and testing and quality assurance monitoring. This will have an overview with a progression bar showing the prototype development stages, and it will include links to detailed prototype reports and testing results. After this, the QA section will be shown with a dashboard monitoring all ongoing metrics for assurance. This includes statistics like response rates, client list completeness, and the production status of the new pumps. Finally, the last section included in our user interface will be a navigation and accessibility region. This is essentially just to ensure that this system will be willing to work and be consistent across multiple forms of media or devices. For example, this will make sure that the interface looks the same way on an iPhone that it would look on an android phone.

The Recall Management System is designed in-house to meet our companies' exact specifications. The automated recall management system (RMS) will identify the location of every impacted pump and deliver an automated message to every customer to take the following actions (reference https://www.fda.gov/medical-devices/medical-device-recalls/becton-dickinson-bdcarefusion-303-recalls-alaris-infusion-pumps-due-compatibility-issues-cardinal):

Stop use of Cardinal Health branded Monoject syringes with the BD Alaris Syringe and PCA Modules.
Replace the current BD Alaris Syringe and PCA Module Compatibility Lists with the updated compatibility lists attached to the letter.
Complete the Customer Response Form that was attached to the letter and return to the BD contact noted on the form to acknowledge receipt of this notification.

Use Case Description: Administering Medication via a Medical Infusion Pump

Primary Actor: Nurse/ Healthcare Provider

Importance Level: High

Secondary Actors: Patient, Electronic Health Record (EHR) System, Pharmacy System

Use case Type: Overview, Essential

Stakeholders and Interests:

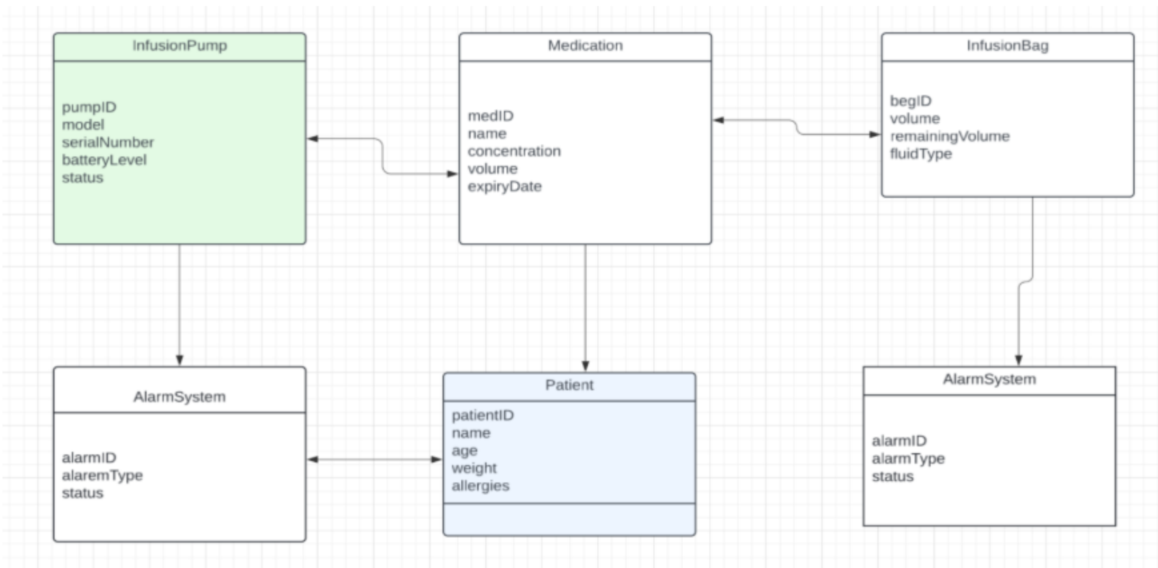
Nurse/ Healthcare Provider: Wants to administer medicine safely and efficiently.
Patient: Wants to receive accurate medication dosing with minimal discomfort.
Hospital Administration: Wants to ensure compliance with safety standards and improve patient outcomes.

Brief Description: This use case describes the infusion pump system administering the correct amount of medicine and/or supplements in a safe way.

Use Case Diagram:


System
Login
Select Medication
Verify Prepartion
Attach Infusion Bag
Start Infusion
Monitor Infusion
Logout
identify
Start Infusion
Monitor Infsuion
Address Issues (if any)
Complete Infusion
Nurse
Patient

Class Diagram:



Test Plan:

Use Case 1	Test case	Expected results
Customers are receiving recall notifications.	Ensuring that the "Automated Recall Management Systems" is sending recall notifications and identifying impacted pumps to updated compatibility list.	Every customer receives a recall notification and identify which pumps were affected and update the compatibility list.
Use Case	Test Case	Expected Results
Customers are receiving recall notifications via email	Ensuring the ARMS emails at least 50 customers per minute	Every 50 customer receives a recall email

		from the ARMS within a minute
Use Case	Test Case	Expected Results
Customers are receiving recall notifications via text message	Select every customer that signed up to receive a email notification	Every customer receives a text from the ARMS system.
Use Case	Test Case	Expected Results
Tracking system makes sure every customer has received a notification and which customers are missed.	Select a customer and see if they did/didn't fill out a response form via email. If they didn't the system alerts the recall team.	Every customer that didn't fill out a response form, the system will notify the recall team.
Use Case	Test Case	Expected Results
Use Case	Test Case	Expected Results
External quality agreement system asses the external product design and changes.	Select any design and the system will notify the System admin and supplier quality team if it is compatible.	Teams are alerted by the system when the design is compatible.
Use Case	Test Case	Expected Results
External quality agreement system asses the external product design and changes.	Select a external brand to determine if they have a compatible design or a non-compatible design will trigger a alert in the (eQAS) and send a notification to the team.	Non-compatible designs will be stopped for production and alerted by the teams

Conversion Strategy:

During the conversion staged we will be completing a parallel conversion strategy. This will allow hospitals with working equipment to continue operating until we can replace all units.

Updating Existing Database for New Pumps (1 week)

Development will work as hard as possible to allow the current system room for the new pumps to be sent out. This should not be a hard task as the system has scalability and was set up for future sales.

Recall Stage (1 Week to 2 Weeks)

In the first stages, the recall team will push the recall notifications to all hospitals and required parties. The recall team will have a small number of employees checking the recalls sent out against a list of all clients to ensure the recall has reached everyone. The coders will also be making sure the ARMS is emailing at least 50 customers a minute to make sure the entire client list is completed as fast as possible. For customers that have opted into the text option, a text will be sent directly to them including a link they can click to read more about the recall. Another small number of employees will be set up to receive the responses and log them into the system when clients fill out the response form. Hopefully all clients can respond within a week and by the end of two weeks all affected pumps will be returned.

Brainstorming/Research Phase (1 to 2 Weeks)

Simultaneously, the entire Research and Development department and several outside sources will begin to break the pumps down and try to get an idea of what is creating the problems with the pumps. This will take 1 to 3 days hopefully. After the issue has been found, the rest of the time will be spent figuring out a full solution to the problem.

Prototyping (1 week)

The engineers and R&D will begin taking the notes from the last phase and building a single prototype that fixes the issues presented. This prototype will then be added to the equipment to make sure the pumps are working correctly. If all is working correctly, we will begin to move to the production phase.

Production (3 weeks)

The production team will begin making all the pump replacements. The production team will work overtime to ensure the pumps are completed as soon as possible.

Testing (1 to 3 Days)

After production has been completed, random pumps will be pulled to ensure a random sample batch and then be tested at several different hospitals. If all works properly, the full roll out will begin.

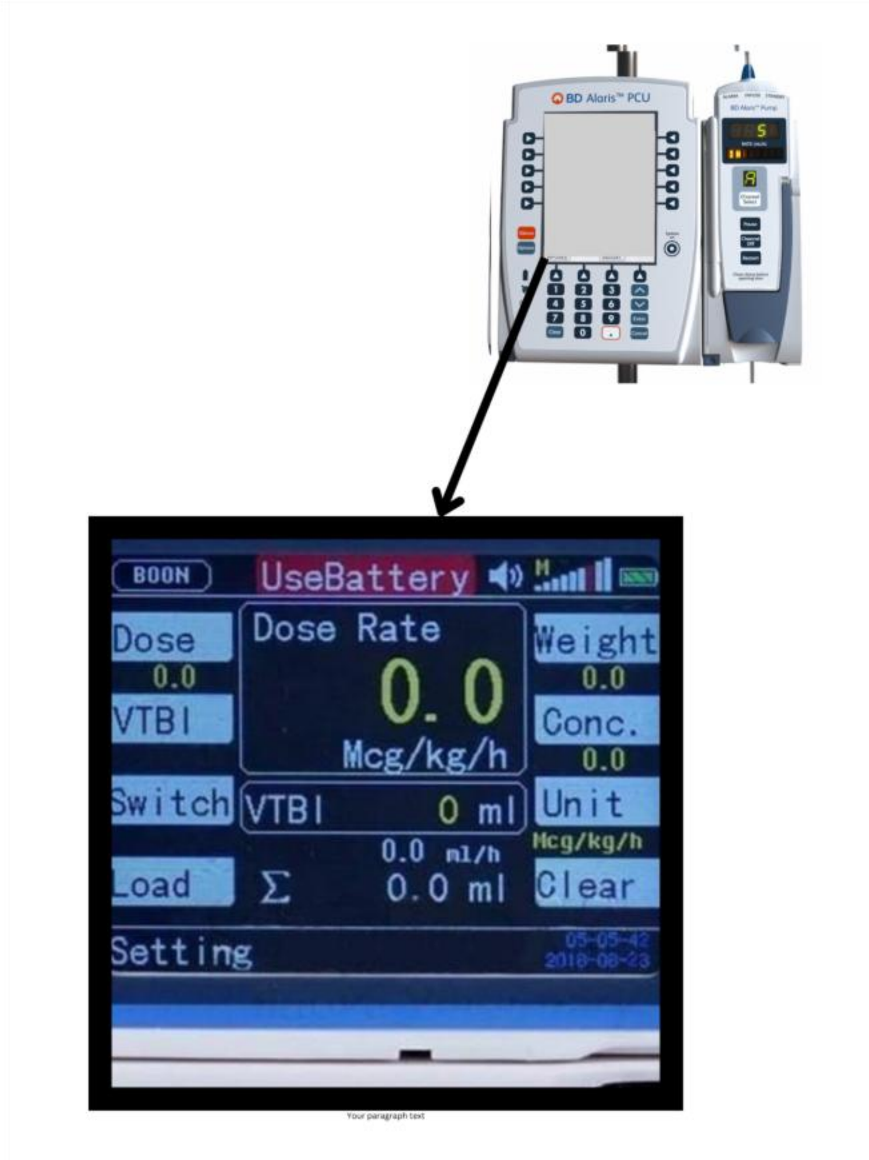
Full Roll Out (2 Weeks)

Replacements for all units will be sent out to clients. A notification will be sent out through the ARMS that instructs clients to notify us when they receive their new units. It will also give them instructions on how to dispose of the defective systems.

Monitoring Quality Assurance (Ongoing)

Going forward all Quality Assurance will be measured through the existing system. Teams will monitor the client list to make sure all clients have received up dated parts. The team will use the ARMS to continue to alert clients who have not responded to the recall or any other steps of the process. The old pumps production will be halted.

User Interface Prototype:



Prototype: Designing a prototype for an infusion pump involves multiple considerations, including hardware, software, and user interface elements. Below is an outline for a systems design of an infusion pump, covering the main components and their interactions.

Hardware
Software
User Interface (UI)
Safety Mechanisms

Microcontroller: The brain of the pump, responsible for controlling all operations.
Motor and Pump Mechanism: Controls the flow rate of the infusion.
Sensors: For monitoring fluid levels, pressure, and detecting occlusions.
Power Supply: Battery and/or AC power.
Display Screen: For user interaction.
Keypad/Touch Interface: For inputting settings.
Alarms and Indicators: For notifying users of errors or conditions requiring attention.

[Power Supply] → [Microcontroller] → [Motor Driver] → [Pump Mechanism]

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[Display Screen] [Flow Sensors] [Pressure Sensors]

↓

[Keypad/Touch Interface]

↓

[Alarm System]

The infusion pump's system design centers around a Microcontroller, which manages all operations, powered by a Power Supply (battery/AC). It controls the Motor Driver and Pump Mechanism for fluid delivery, interfacing with Sensors for flow, pressure, and fluid levels. The Display Screen and Keypad/Touch Interface form the user interface, allowing parameter settings and status monitoring. The Alarm System provides alerts for detected issues.

The Automated Recall Management System (ARMS) is a critical component designed to enhance safety, reliability, and efficiency of the infusion pump. ARMS continuously monitors the pump’s operation, ensuring accurate delivery of fluids and immediate detection of any anomalies.

The Recall Management System is important for managing the process of recalling infusion pumps in case of a discovered fault or safety issue. ARMS ensures that faulty devices are quickly identified, tracked, and rectified, to minimize patient risks and enhance safety.

ARMS is a vital component of the infusion pump’s design, ensuring that any issues are swiftly addressed to maintain patient safety.